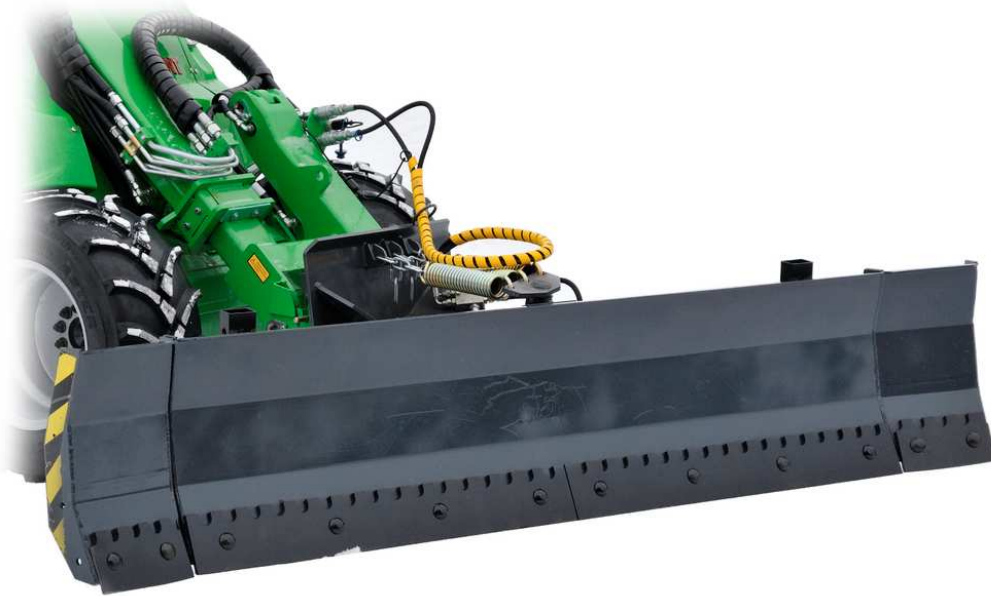


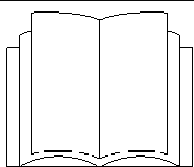


Operator's Manual for Attachment



Dozer blade

Product number:	300-700-series loaders:	A2867 A2944 A410365
	200-series loaders:	A21093



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

Manufacturer:

AVANT
AVANT TECNO OY
e-mail: sales@avanttecno.com

Ylötie 1
33470 Ylöjärvi
FINLAND
Tel. +358 3 347 8800
Fax +358 3 348 5511

www.avanttecno.com

CONTENTS

1.	Foreword	3
2.	Designed purpose of use	3
3.	Safety instructions for using the Dozer Blade	3
4.	Dozer Blade specifications	5
5.	Assembly of the Dozer Blade	6
5.1.	Mounting the Dozer Blade to the loader	6
5.2.	Connecting and disconnecting the hydraulic fittings	7
6.	Instructions for use	8
6.1.	The edge of the Dozer Blade	9
6.2.	Blade edge spring release	9
7.	Maintenance and service	9
7.1.	Cleaning and lubrication	10
8.	Warranty terms	10

Appendix 1: EC Declaration of Conformity Dozer blade models A2867, A2944, A21093

Appendix 2: EC Declaration of Conformity Dozer blade model A410365

This manual uses the following warning symbols to indicate factors that must be taken into account to reduce the risk of personal injury or property damage:

	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or property damage. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.
---	--



This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment, particularly the use of the equipment on road traffic. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this Avant attachment. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

With these instructions, even an inexperienced user can use the attachment and loader safely. The manual includes important instructions for experienced Avant operators as well. Ensure that all persons using the loader have received proper guidance and familiarised themselves with the manual of the loader and attachment and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost or damaged, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

This AVANT Dozer Blades are attachments suitable for use with Avant compact loaders for ploughing snow and also for ploughing soil or sand. All dozer blade models are equipped with hydraulic slewing, which enables to choose the side to which the material will be transferred without interrupting work. All dozer blade models are also equipped with a spring release mechanism which protects the blade and driver in case an obstacle is hit.

The two solid support legs on the dozer blade models A2944 (2 000 mm) and A410365 (2 500 mm) reduce blade wear and improve manoeuvrability. These models also have two 250 mm extensions as standard. The blade extensions can be mounted so that the blade is straight or so that they are tilted forward, so as the blade will collect the material being ploughed. The wider dozer blades are also higher to enable ploughing of larger amounts of material at once. The wider models are also equipped with a turnable blade with straight / ice cutting edge. The wide models suit well with the larger loader models.

The attachments have been designed to require as little maintenance as possible. Regular maintenance tasks can be performed by the operator. Appropriate protective equipment and original spare parts must be used for the maintenance. Demanding repair and maintenance operations are to be left for professional maintenance. Familiarise yourself with the manual's instructions regarding service and maintenance. Please contact your Avant retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for using the Dozer Blade

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use or maintenance may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards.

The purpose of these instructions is to help you to:

- use the equipment in a safe and efficient manner
- observe and prevent any hazardous situations
- keep the equipment intact and ensure a long service life



- When mounting the dozer blade to the loader, ensure that the locking pins of the loader's quick attach plate are **in the lower position** and that they have **locked** the attachment **to the loader**.



- Do not drive at high speed unless you can be sure that there are no obstacles at the working area. The loader may stop suddenly if it hits an obstacle, regardless of the spring release mechanism. Remember that there may be hidden obstructions or fragile surfaces under the snow.
- Pay attention to the surroundings and any other persons and machines moving in the vicinity. Pay attention to the contours of the terrain as well as other hazards, such as branches and trees reaching the driver's area. On slippery areas drive calmly and deliberately.
- Shut down the loader engine and release residual hydraulic pressure before adjustments or maintenance work. Read instructions concerning maintenance of the equipment.
- Do not modify the structure of the attachment or the loader in a way that would affect safety.
- Also read the safety instructions for the loader in the loader manual.



Note that the loader boom moves down and may be tilted even if the engine has been stopped. Never go under a raised boom or dozer blade component.



Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine. Note that driving characteristics of the loader may be altered by the load at the front and on slippery surfaces.



Observe the local regulations concerning the use of the equipment on public roads. The loader may need to be registered for road use. Contact dealer for more information.

Remember to wear proper protective clothing:



- The noise level at the driver's seat may exceed 85 dB(A) depending on the loader model and working conditions. Wear hearing protection while working with the loader.



- The use of safety boots is recommended while working with the loader.
- Wear protective gloves when handling hydraulic hoses.

4. Dozer Blade specifications

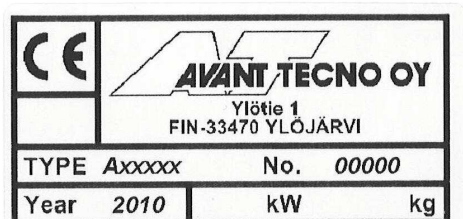
Table 1 – Dozer Blade specifications

Product name:	Dozer Blade 1400	Dozer Blade 1400	Dozer Blade 2000	Dozer Blade 2500
Product number:	A21093	A2867	A2944	A410365
Working width:	1400 mm	1400 mm	1500 + 500 mm (2 x 250 mm)	2000 + 500 mm (2 x 250 mm)
Total weight:	100 kg	100 kg	165 kg	260 kg
Blade height:	380 mm	380 mm	500 mm	520 mm
Slewing angle:	±30°	±30°	±30°	±30°
Turnable blade with straight / ice cutting edge:	-	-	x	x
Support legs:	-	-	x	x
Available optional accessories:	Rubber Blade 1400 mm A32383 Ice Blade 1400 mm A32315 Blade Extensions 2 x 180 mm A48869		Rubber Blade A33833	-
Maximum allowed input of hydraulic energy:	22,5 MPa (225 bar) 70 l/min			
Suitable Avant loader	216, 218, 220	300-600 (700)	(300) 400-700	(400-500) 600-700

Listed below are the labels and markings, which must be visible on the equipment. Replace any warning label, that has become unclear or has detached completely. New labels are available via your retailer or contact information provided on the cover.



The warning labels contain important safety information. Replace damaged or missing warning labels with new ones.



Identification plate of the attachment

- A46771 – Read instructions before use.
- A46772 – Do not go under a raised attachment, stay away from raised equipment.
- A46803 – Crushing hazard – Stay clear of moving components
- Hazard warning tape at corners (model A410365 only)



A46771



A46772



A46803

5. Assembly of the Dozer Blade

The dozer blades are delivered in working condition. Mount the attachment and its hydraulic hoses to the loader as instructed and begin the practise of use at a safe area. The hydraulic cylinder and the hydraulic hoses may have air trapped inside them, which causes imprecise movement of the blade at the first use. The hydraulic system will bleed itself as it is used; no separate bleeding procedure is required.

These instructions are applicable to mounting the attachment to the boom quick attach plate of Avant loaders that have standard quick attach locking. If your machine is equipped with a hydraulic locking and/or serial quick attach system, refer to the relevant manuals for additional mounting instructions.

5.1. Mounting the Dozer Blade to the loader

Mounting the attachment to the loader is quick and easy, but it must be done carefully. An attachment that has not been locked may come off the loader and cause hazardous situations. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised to a height of over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3 and the instructions provided in the loader manual.

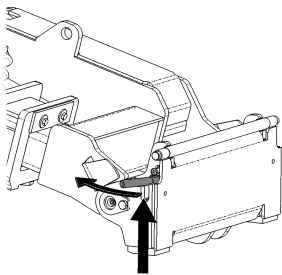
The attachment is mounted to the loader boom by using the quick attach plate and the counterpart on the attachment. The locking is semi-automatic; when the boom front plate is tilted towards the loader; the locking pins are released as they hit the pin guides on 300-700 series loaders or the boom on 200 series loaders.



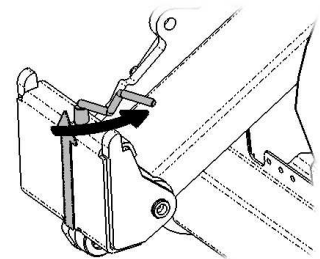
Never move or raise an attachment that has not been locked.

The dozer blade is mounted to the loader as follows:

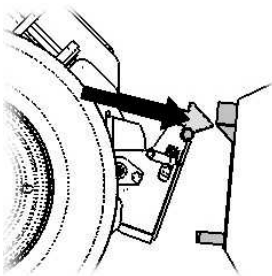
Step 1:



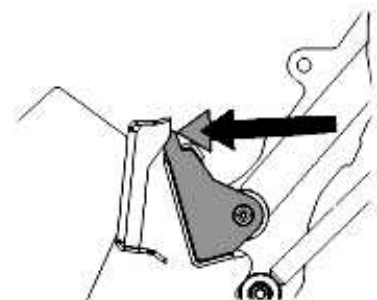
- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- Ensure that the hydraulic hoses are not in the way during installation.



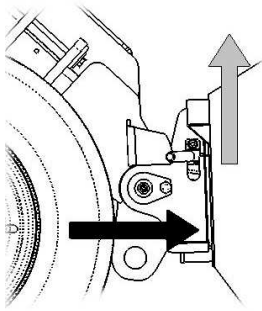
Step 2:



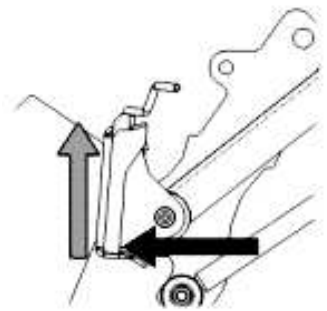
- Turn the quick attach plate hydraulically to an obliquely forward position. The angle between the loader front plate and attachment coupling should be about as shown in the adjacent picture.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this. If the dozer blade is on ground level, it may be necessary to lift the attachment blade by hand a little.



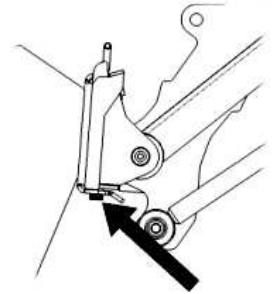
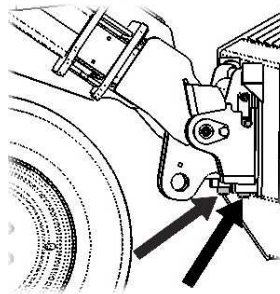
Step 3:



- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Turn the boom control lever further left so that the automatic locking of the attachment triggers the locking pins.



Before moving or raising the attachment, ensure that the locking pins are in the lower position and come through the fasteners on both sides. Unlocked attachment may fall on the boom or operator and cause injuries.



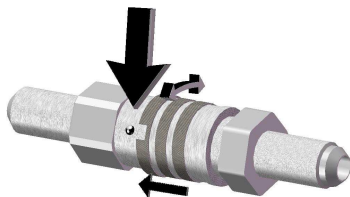
Do not use the automatic locking of the locking pins **at a height of over a metre!** If the locking pins do not return normally to the locking position when the front plate is tilted, do not raise the attachment any more. Lower the attachment on the ground and secure the locking manually.

5.2. Connecting and disconnecting the hydraulic fittings

Before connecting or disconnecting fittings, shut down the loader engine and move the working hydraulics lever on the loader to its extreme positions a few times. This way, there is no remaining pressure in the system and the fittings are easy to connect.

The two hydraulic fittings of the dozer blade are connected to the corresponding fittings on the loader. To connect and disconnect the fittings, move the collar at the end of the female fitting. The hoses should be connected to the loader by connecting the fittings equipped with coloured caps to each other, so that the blade slewing works the same way each time it is used. This way the operating the blade slewing does not cause unexpected movements.

In a 200-series machine, the hose can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their locking. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The fitting collar may also turn by accident into the locking position, making it impossible to connect or disconnect the fittings.



In the 200-series equipment, there is a small groove in the collar of all female fittings and a small ball below the collar, as illustrated in the adjacent picture. When the fittings need to be disconnected, the above-mentioned groove and ball need to be aligned. Then, the grooved collar of the female fitting is pulled to disconnect the fittings from each other.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure. High-pressure fluid ejection may cause serious injuries.



When removing the attachment, always disconnect the hydraulic fittings before detaching the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.

Keep the fittings as clean as possible. Use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. Never leave the hoses hanging on the ground; place them on top of the equipment.

6. Instructions for use

After mounting the dozer blade as described in chapter 5 it is ready to be used. Check the attachment and the operating environment once more before starting the work. Quick inspection of the equipment and the operating area prior use is part of ensuring safety and the best performance of the equipment.

- Check that all obstacles, including any possibly hidden ones, have been removed from the working area before operation. Remember correct and healthy working methods and avoid getting down unnecessarily from the loader.
- Check that the spring release mechanism works as intended and that all springs are in place. Ensure that there are no obstructions near the slewing hinge and that the hydraulic hoses cannot get squeezed between any components or that they are not exposed to stretch at any position.
- Check regularly to ensure there are no hydraulic leaks.
- Ensure that the loader and the attachment are used in a safe and intended manner. Operate the attachment from the driver's seat only. Ensure also that the operating area is illuminated well enough for safe use of the equipment.



If your loader is equipped with a seat belt, use it when working with dozer blade. The loader may stop suddenly and unexpectedly when an obstacle is hit

The vertical position of the blade will affect the outcome and accuracy of dozing. As a rule, the blade should be adjusted to an upright position. According to circumstances you may try to tilt the blade a little to front or back for example when removing ice. The blade should not be pushed hard against the ground as this will affect the handling characteristics of the loader.

If your loader is equipped with the optional boom floating system, it is advisable to switch it on. The floating system helps to achieve more even outcome. More information of the boom floating is shown in the operator's manual of the loader.

Check the operating area before using the equipment:



- Make sure, that there are no hidden obstacles, for example, under the snow. Running into an obstacle may stop the machine suddenly and unexpectedly regardless of the spring tilting system or the boom floating of the loader.
- Ensure that there is nobody else in the working area. Do not allow assisting persons to go directly in front of the dozer blade or to reach the boom danger zone. Also ensure that there is nobody in the way when reversing with the loader.

6.1. The edge of the Dozer Blade

Adjust the cutting edge of the dozer blade so that only the replaceable edge plate touches the ground. The frame of the dozer blade must not touch the ground surface in order to prevent it from wearing. Replace the cutting edge of the dozer blade before the frame comes to contact with the ground. The wider dozer blade models have a straight / ice cutting edge as standard and rubber blade is available for some models. Tighten the bolts of the cutting edge evenly.

The blade extensions on the 2 000 mm and 2 500 mm wide dozer blades are different on the left and the right side of the blade. The extensions may be mounted to a straight or to a collecting position.

6.2. Blade edge spring release

The dozer blades are equipped with a hinged mechanism with spring load, which allows the blade to tilt backwards if an obstacle has been hit. The tilting also protects the blade edge. Pretensioned springs hold the blade in upright position. Dozer blade 2 or 3 springs depending on the model of the dozer blade. Make sure that the tilting mechanism works as intended. If necessary, the springs may be tightened, if heavier materials are to be dozed. Grasp on chain at a time with a secure grip by both hands and adjust the chains with care.

7. Maintenance and service

Continuous maintenance of the dozer blade includes regular cleaning and lubrication of the equipment. The dozer blades have 6 lubricating points which can be lubricated with a standard grease press.

Check the condition of the hydraulic hoses and components when the engine has been turned off and the residual hydraulic pressure has been relieved. Do not use the equipment, if you have discovered a fault in the hydraulic system. Leaking hydraulic fluid may penetrate the skin and cause serious injuries. Hydraulic fluid is also harmful to the environment.

Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of any hose has worn off. If there are signs of leaking, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component in question must be replaced. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Spare parts are available from your nearest AVANT retailer or service point.



Never tighten a hydraulic fitting under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment if you have discovered a fault in the hydraulic system.



7.1. Cleaning and lubrication

You may use a pressure washer and a mild detergent to wash the equipment. Avoid spraying directly at the hydraulic components or using solvents when washing. In addition, mind the labels on the attachment. Lubricate the lubrication points after washing. Lubrication points are located at the both ends of the hydraulic cylinder (2), the blade tilting mechanism (2) and the blade slewing joint (2). All lubrication nipples are standard R1/8" nipples. Replace any damaged nipples.



Lubricant must be added about 1 to 3 times a year, depending on usage and operating conditions. Clean the end of the nipple before greasing. Only add the grease amount of a couple of grease press pushes at a time.

Clean the attachment thoroughly before long storage periods, touch up the paint where needed to prevent rust damage and lubricate the lubricating points. Loosen the springs of the tilting mechanism to ensure long service life of the springs. The dozer blades should be stored in a position where the cylinder is completely retracted. Protect the visible part of the cylinder shaft by applying a small amount of oil to it. Do not store the attachment directly against the ground; place it on blocks of wood or on a pallet, for example, and ensure the stability of the equipment.

8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) from the date of purchase for the attachment it manufactures.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:

AVANT TECNO OY

Osoite / Adress / Address / Adresse:

Ylötie 1

33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
 SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:
 SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
 SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:
 SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

Kuvaus: Avant-kuormainten kanssa käytettävä puskulevy
 Beskrivning: Schaktblad; arbetsredskap för Avant lastare
 Description: Dozer blade; attachment for Avant loaders
 Beschreibung: Planierschild; Anbaugerät für Avant Radlader

Mallit / Modeller / Models / Modellen:

Avant		200 series	300- 700 series
Puskulevy	140 cm	A21093	A2867
Schaktblad			
Dozer blade	200 cm	-	A2944
Planierschild			

Allekirjoitus / Namnteckning / Signature / Unterschrift: _____ 9.12.2011 Ylöjärvi, Finland

Risto Käkelä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer



EY-vaatimustenmukaisuusvakuutus EC Declaration of Conformity

Valmistaja / Manufacturer : AS SAMI
Osoite / Address: Tule 20
76505 Saue, HARJUMAA, ESTONIA
www.sami.ee

Vakuutamme täten, että alla mainittu tuote täyttää konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
EN ISO 12100, EN ISO 4413

We hereby declare that the product listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
EN ISO 12100, EN ISO 4413

Kuvaus: Avant-kuormainten kanssa käytettävä puskulevy 2500
Description: Dozer Blade 2500; attachment for Avant loaders

Allekirjoitus / Signature :  09.12 2011 Saue, Estonia

Gert Jugala,
Johtaja/
Director





AVANT[®]